Set Exercise

A university stores courses by course code. Each course has a set of registered student names. A student must not appear twice in the same course. Build a simple console application using Dictionary<string, HashSet<string>>.

Tasks

1. Create courses

Create a Dictionary<string, HashSet<string>> called courses. Add OOP with Peter and Sara; add Database with Henrik and Emma.

2. Register students

Add Mads to OOP. Try adding Peter to OOP again. Print whether the second Add() succeeded and the total number of OOP students.

3. Find a student

Use ContainsKey() to check that Database exists, then Contains() to check whether Emma is registered. Print a confirmation when both are true.

4. Display registrations

Use nested foreach loops to print each course code and the students registered in that course.

Starter Code

```
using System;

using System.Collections.Generic;



public class Program

{

    public static void Main(string[] args)

    {

        // TODO 1: Create a Dictionary of courses and student sets

        // TODO 2: Add OOP and Database courses

        // TODO 3: Register Mads and try adding Peter again

        // TODO 4: Check for Database and Emma

        // TODO 5: Display each course and its students

    }

}
```

Expected Output

```
Peter added again: False

OOP students: 3

Emma is registered for Database

Course: OOP

Peter

Sara

Mads

Course: Database

Henrik

Emma
```

Note: HashSet does not guarantee the order in which student names are displayed.